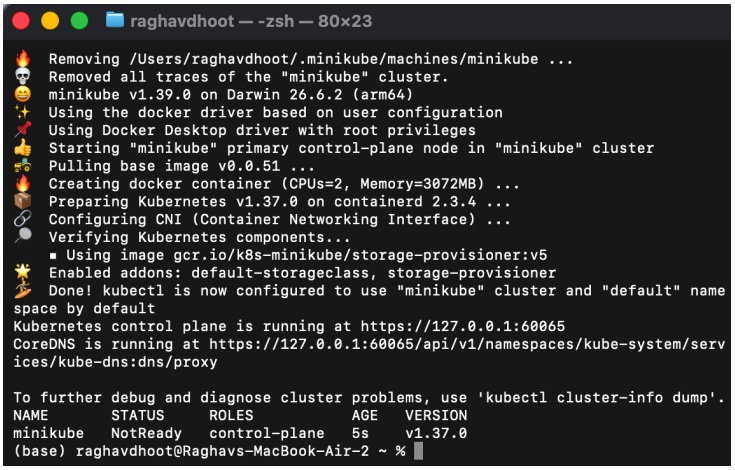
SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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॥वसुधैव कुटुम्बकम्॥

Department of Computer Science and Engineering
DevOps Lab - L3

Lab Assignment No: 12	
Name of Student	Raghav Dhoot
PRN	23070122170
Batch	2023 - 2027
Class	CSE - B1
Academic Year & Semester	2026 - 2027, 7th Semester
Title of Assignment	Kubernetes Pod & Deployment
Implementation & Theory	 Local cluster setup — Hit a cri-dockerd.sock permission error on first Minikube start; resolved by running minikube delete and restarting fresh with minikube start --driver=docker. Verified with kubectl cluster-info and kubectl get nodes, confirming a healthy single-node cluster.

```
(base) raghavdhoot@Raghavs-MacBook-Air-2 my-flask-app % cd ~/devops-a10/my-flask-app
docker build -t my-flask-app:v2 .
docker images | grep my-flask-app
[+] Building 14.0s (11/11) FINISHED                                 docker:default
 => [internal] load build definition from Dockerfile                        0.0s
 => => transferring dockerfile: 183B                                        0.0s
 => [internal] load metadata for docker.io/library/python:3.11-slim         6.0s
 => [auth] library/python:pull token for registry-1.docker.io               0.0s
 => [internal] load .dockerignore                                           0.0s
 => => transferring context: 2B                                             0.0s
 => [1/5] FROM docker.io/library/python:3.11-slim@sha256:e88e9763f943ec18   4.9s
 => => resolve docker.io/library/python:3.11-slim@sha256:e88e9763f943ec18   0.0s
 => => sha256:73a63ac3ec4a2ac51ab19307c2a807cd83546a504 14.41MB / 14.41MB   2.8s
 => => sha256:e88e9763f943ec1834f992a4b51e0f24500486803 10.37kB / 10.37kB   0.0s
 => => sha256:1847ca1d1bc792b4b58fff39c82c67942112e3e1ebc 1.75kB / 1.75kB   0.0s
 => => sha256:8129910271f9764b5c05038d351ad7b9f9d94b85351 5.50kB / 5.50kB   0.0s
 => => sha256:bbeda6b4abb749f743f4547ef5a28070a14de8cae 30.20MB / 30.20MB   2.4s
 => => sha256:9eb0fb93ee0808559f67707773b41df45d2dd34a5b4 1.28MB / 1.28MB   0.8s
 => => sha256:7ca0c331ecfd0200dc735b8a2ac2be32f9c43d5ea0945cb 250B / 250B   1.3s
 => => extracting sha256:bbeda6b4abb749f743f4547ef5a28070a14de8caef211daa   1.1s
 => => extracting sha256:9eb0fb93ee0808559f67707773b41df45d2dd34a5b4aec77   0.1s
 => => extracting sha256:73a63ac3ec4a2ac51ab19307c2a807cd83546a50407c2c52   1.1s
 => => extracting sha256:7ca0c331ecfd0200dc735b8a2ac2be32f9c43d5ea0945cbb   0.0s
 => [internal] load build context                                           0.0s
 => => transferring context: 1.04kB                                         0.0s
 => [2/5] WORKDIR /app                                                      0.2s
 => [3/5] COPY requirements.txt .                                           0.0s
 => [4/5] RUN pip install --no-cache-dir -r requirements.txt                2.7s
 => [5/5] COPY . .                                                          0.0s
 => exporting to image                                                      0.1s
 => => exporting layers                                                     0.1s
 => => writing image sha256:cbd5221e7eb2c8ba6b1356c6952c294dc31f70e414b6e   0.0s
 => => naming to docker.io/library/my-flask-app:v2                          0.0s
my-flask-app:v2                                cbd5221e7eb2       166MB             0B
(base) raghavdhoot@Raghavs-MacBook-Air-2 my-flask-app %
```

Image accessibility — Since Minikube runs its own isolated Docker daemon, pointed the local shell at it with eval $(minikube docker-env). Confirmed my-flask-app wasn't present in that daemon's image list, then rebuilt the image (my-flask-app:v2) directly inside the Minikube Docker context so the cluster could pull it locally without needing a registry.

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: web-app-deployment
  labels:
    app: web-app
spec:
  replicas: 2
  selector:
    matchLabels:
      app: web-app
  template:
    metadata:
      labels:
        app: web-app
    spec:
      containers:
      - name: web-app-container
        image: my-flask-app:v2
        ports:
        - containerPort: 5000
        imagePullPolicy: IfNotPresent
```

Deployment YAML created — Wrote web-deployment.yaml defining a Deployment with 2 replicas, using label selectors (app: web-app) to link the Deployment to its Pods, referencing the locally built my-flask-app:v2 image with imagePullPolicy: IfNotPresent.

```
deployment.apps/web-app-deployment created
(base) raghavdhoot@Raghavs-MacBook-Air-2 k8s-manifests %
```

Applied the manifest — kubectl apply -f web-deployment.yaml created the Deployment successfully.

```
NAME                                 READY   STATUS    RESTARTS   AGE
web-app-deployment-8b54cd854-cvcwf   1/1     Running   0          34s
web-app-deployment-8b54cd854-wc6sc   1/1     Running   0          33s
(base) raghavdhoot@Raghavs-MacBook-Air-2 k8s-manifests %
```

Verified Pods running — kubectl get pods showed both replica Pods (web-app-deployment-*) in Running state with 1/1 containers ready, confirming the Deployment successfully scheduled and started the application across multiple Pods.